Problem 6.13

Suppose that your sister repays you \$62 two months after she borrows \$60 from you. What effective annual rate of interest have you earned on the loan? Assume compound interest.

Let the effective annual interest rate be i. Since two months is $\frac{2}{12} = \frac{1}{6}$ of a year, we have

$$62 = 60(1+i)^{1/6}.$$

Dividing by 60,

$$\frac{62}{60} = (1+i)^{1/6}.$$

Raising both sides to the 6th power,

$$1 + i = \left(\frac{62}{60}\right)^6.$$

Therefore,

$$i = \left(\frac{62}{60}\right)^6 - 1.$$

Hence,

$$\boxed{i = \left(\frac{62}{60}\right)^6 - 1 \approx 0.2165}.$$

So the effective annual interest rate is approximately

$$\boxed{21.65\%}.$$